An evaluation of mercury concentrations in three brands of canned tuna.

There is widespread concern over the presence of Hg in fish consumed by humans. While studies have been focused on determining the Hg concentration in sport fish and some commercial fish, little attention has been directed to canned tuna; it is widely held that concentrations are low. In the present study, the amount of Hg present in canned tuna purchased in Las Vegas, Nevada, USA, was examined, and the brand, temporal variation, type, and packaging medium impacts on Hg concentrations in tuna were explored. A significant ($p < 0.001$) brand difference was noted: Brand 3 contained higher Hg concentrations ($\bar x$ standard deviation (SD) (0.777 +/- 0.320 ppm) than Brands 1 (0.541 +/- 0.114 ppm) and 2 (0.550 +/- 0.199 ppm). Chunk white tuna (0.619 +/- 0.212 ppm) and solid white tuna (0.576 +/- 0.178 ppm) were both significantly ($p < 0.001$) higher in mean Hg than chunk light tuna (0.137 +/- 0.063 ppm). No significant temporal variation was noted, and packaging had no significant effect on Hg concentration. In total, 55% of all tuna examined was above the U.S. Environmental Protection Agency's (U.S. EPA) safety level for human consumption (0.5 ppm), and 5% of the tuna exceeded the action level established by the U.S. Food and Drug Administration (U.S. FDA) (1.0 ppm). These results indicate that stricter regulation of the canned tuna industry is necessary to ensure the safety of sensitive populations such as pregnant women, infants, and children. According to the U.S. EPA reference dose of 0.1 microg/kg body weight per day and a mean Hg value of 0.619 ppm, a 25-kg child may consume a meal (75 g) of canned chunk white tuna only once every 18.6 d. Continued monitoring of the industry and efforts to reduce Hg concentrations in canned tuna are recommended. Environ. Copyright 2009 SETAC.